

WEEK 2 – IDENTIFYING DATA REQUIREMENTS

Online Shopping Database

[1. Introduction]

In an online shopping system, a database is required to store and manage information about customers, products, orders, payments, shipments, suppliers, and reviews. Each entity contains specific attributes, and every entity should have a primary key to uniquely identify each record.

[2. Entities and Attributes]

1. Customer – Customer_ID (PK), Name, Email, Phone, Password, Address, City, Pincode.
2. Product – Product_ID (PK), Product_Name, Description, Price, Stock_Quantity, Category_ID (FK), Supplier_ID (FK).
3. Category – Category_ID (PK), Category_Name, Description.
4. Supplier – Supplier_ID (PK), Supplier_Name, Email, Phone, Address.
5. Order – Order_ID (PK), Customer_ID (FK), Order_Date, Total_Amount, Order_Status, Delivery_Address.
6. Payment – Payment_ID (PK), Order_ID (FK), Payment_Date, Payment_Method, Amount, Payment_Status, Transaction_ID.
7. Shipment – Shipment_ID (PK), Order_ID (FK), Shipment_Date, Expected_Delivery_Date, Delivery_Date, Shipment_Status, Tracking_Number.
8. Review – Review_ID (PK), Customer_ID (FK), Product_ID (FK), Rating, Review_Text, Review_Date.

[3. Primary Keys]

1. Customer – Customer_ID
2. Product – Product_ID
3. Category – Category_ID
4. Supplier – Supplier_ID
5. Order – Order_ID
6. Payment – Payment_ID
7. Shipment – Shipment_ID
8. Review – Review_ID

[4. Foreign Keys]

1. Product.Category_ID → Category.Category_ID
2. Product.Supplier_ID → Supplier.Supplier_ID
3. Order.Customer_ID → Customer.Customer_ID
4. Payment.Order_ID → Order.Order_ID
5. Shipment.Order_ID → Order.Order_ID
6. Review.Customer_ID → Customer.Customer_ID
7. Review.Product_ID → Product.Product_ID

[5. Summary]

The identified entities and their attributes form the basic data requirements of the online shopping database. Primary keys uniquely identify records, while foreign keys establish relationships between different entities. This database structure will help the system efficiently manage customers, products, orders, payments, shipments, suppliers, and reviews.